Global Knowledge®

JavaScript

Working with Events

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Event Driven JavaScript

Making your website or app react to events

Event-driven JavaScript

- Events happen –mostly– in the browser
  - either system-dependent (`scroll`, `load`, etc)
  - or initiated by the user (`click`, `mouseover`, `hover`)
  - Also: events thrown by external systems (http-connection, websockets)
- Events are fired by (elements on the) page
- Events are always written in lowercase.

- Events *can* be captured and trigger the execution of a script
  - Capturing is not required
  - If not captured, an event simply dies, without something happening
  - Events 'bubble up' the DOM tree (most of them)
- Some events can be fired by all elements
  - like `mouseover`
- Some by only some elements
  - like `change`

Events example

- Window
  - `load`
  - `resize`
  - `unload`
  - `scroll`
- input, select, textarea
  - `change`
  - `focus`
  - `blur`
- button
  - `focus, blur`
- visible elements
  - `mouseover`
  - `mouseup`
  - `mouseout`
  - `mousedown`
  - `mousemove`
- all
  - `click`
- …dozens of others (countless!)

Relatively new: mobile events

- touch
- tap
- swipe
- pinch
- zoom
- rotate
- Orientationchange
- And more…

Events ARE NOT HTML-attributes

- Don’t do this inline: (*deprecated!*)
  - onload
  - onmouseover
  - etc
  - `<button onclick="doFunction()">`
- Instead, use `addEventListener()`
  - `load`
  - `mouseover`
  - etc.
- Write a function, the Event handler, to do something
  - This can be an anonymous function or named function, like
  - onload()
  - onmouseover() etc.

In short, NEVER do this:

HTML

```html
<button onclick="alert('Hello, this is my old-fashioned event handler!');">
  Press me
</button>
```

Instead, do something like:

HTML

```html
<input id="textBox" type="text" />
<div id="output"></div>
```

Or, use document.getElementById (...)

JS

```js
const textBox = document.querySelector("#textBox");
const output = document.querySelector("#output");
textBox.addEventListener("keydown", (event) => {
  output.textContent = `You pressed "${event.key}".`;
});
```

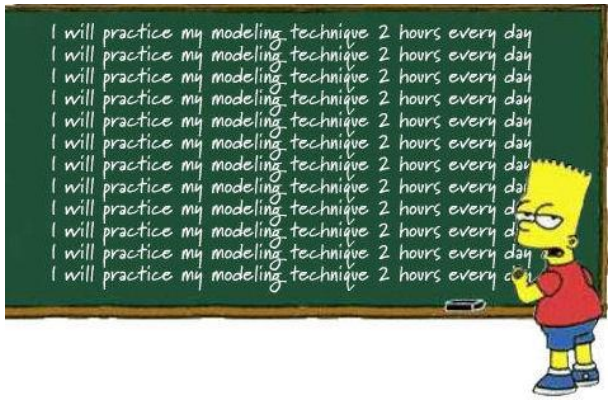
The event is passed as a parameter to the event handling function

Mini workshop

Create the example from the previous slide

- Catch a keypress
- Show in the console, or show in the page which key was pressed

- Q: How can you get an overview of all event-properties available in the `event` object?



Example – getting the cursor location

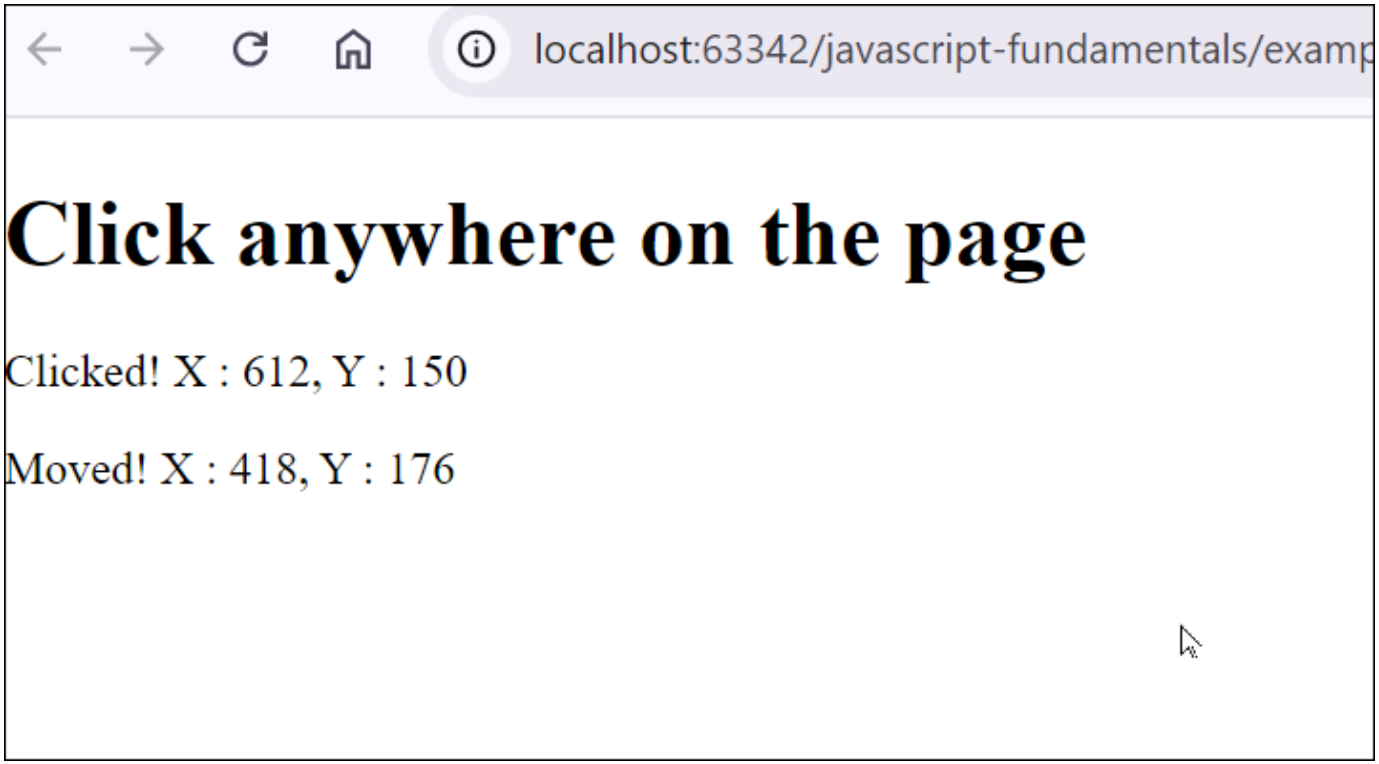
```
<script>
    // example: catching mouse clicks and mouse moves.
    window.onload=function(){
        document.addEventListener('click', handleClick);
        document.addEventListener('mousemove', handleMove);
    };

    // Event handlers
    function handleClick(evt){
        const xCoord = evt.clientX;
        const yCoord = evt.clientY;
        const msg = 'Clicked! X : ' + xCoord + ', Y : ' + yCoord;
        document.getElementById('coords').innerHTML = msg;
    }

    function handleMove(evt){
        const xCoord = evt.clientX;
        const yCoord = evt.clientY;
        const msg = 'Moved! X : ' + xCoord + ', Y : ' + yCoord;
        document.getElementById('mouseMove').innerHTML = msg;
    }
</script>
```

```
<h1>Click anywhere on the page</h1>
<p id="coords"></p>
<p id="mouseMove"></p>
```

Sample output



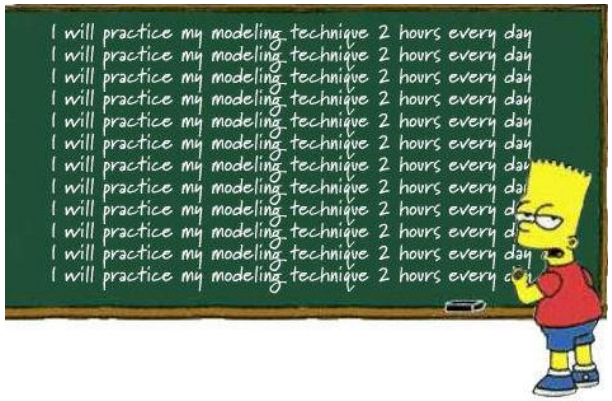
The events are `click` and `mousemove`.

The properties used are `clientX` and `clientY`.

52_mouseEvent.html

Workshop #30

- Create a script where after a button is clicked, all links in the page change background colour.
- Usage:
  - A simple HTML-page, containing some hyperlinks
  - An array of colours
  - `getElementsByTagName()` or `document.querySelectorAll()`
  - Loop over a NodeList object



Sample output

Change link backgrounds on mouse click

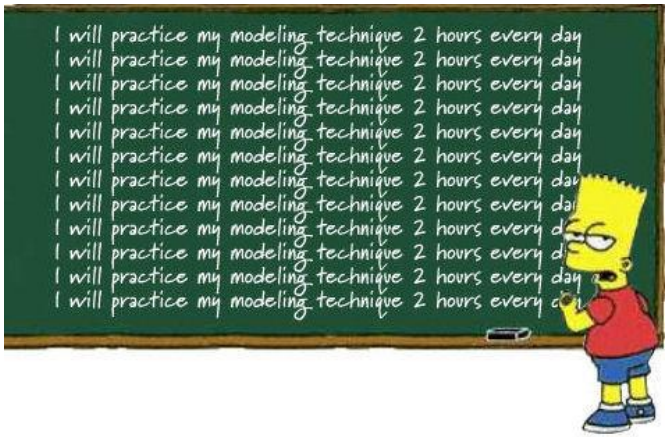
This is for learning to work with JavaScript events

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean in eros urna. Ut posuere ultrices sagitti
libero sed convallis facilisis, lacus quam malesuada tellus, at condimentum felis sem quis ligula. Sed e
porttitor fermentum. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus
commodo, nulla et aliquet ullamcorper, massa nibh viverra enim, sit amet tincidunt velit purus eu ipsu
interdum rhoncus. Sed tellus nunc, rhoncus vel semper vitae, vehicula sit amet nunc. Pellentesque id e
dignissim vitae at justo. Phasellus rutrum sem ac nibh condimentum ac vestibulum justo vehicula. Sed
accumsan libero sollicitudin quis pulvinar tortor aliquet.

Change link backgrounds

Workshop #31

- Write a script that checks a value in a form:
- a `<textarea>` element must contain no more than xx characters. Set this number yourself.
  - (Yes, like Twitter, Mastodon and other Social Media Platforms)
- Show how many characters the visitor can still type
- Show if the number of characters is exceeded.
- the check takes place after each character is typed in the textarea
  - Tip: use the event `input`



Sample output

Character Counter Example

Type in the textarea below. The maximum allowed characters are 100.

```
Lorem ipsum dolor sit,
bladybladybla
```

You can type 64 more characters.

Color (CSS-class) if OK

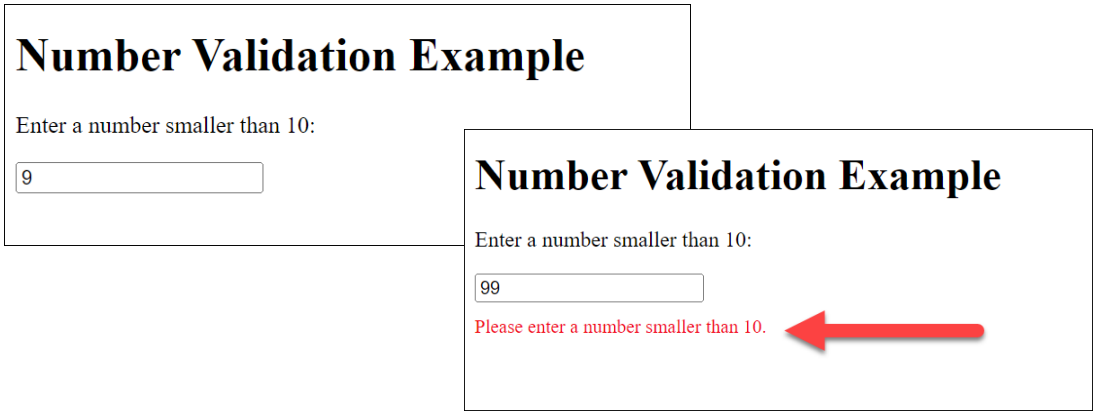
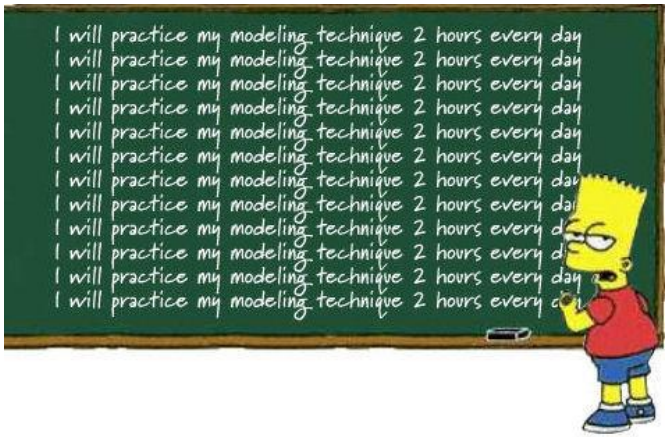
Character Counter Example

Type in the textarea below. The maximum allowed characters are 100.

```
Lorem ipsum dolor sit,
bladybladybla<script>
// Maximum number of characters
allowe const maxChars = 100;
```

You have exceeded the limit by 5 characters!

Color (CSS-class) if Not OK

Workshop #32

- An input field must contain a number smaller than 10
- This check takes place after this input field has lost focus
  - This comes in handy for age checks, not exceeding a certain value, etc.
- This is an easy one!